

The Output Gap Does Not Exist

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Abstract

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1 The Ambiguous Gaps

Motivation. A single number labeled “the output gap” is routinely reported and discussed by academics and policy makers alike. Yet the implications of a value like +2% are not clear: does it imply inflationary pressure, some over-utilization of resources, or maybe just that the economy is in a very nice period and will slowly converge back to normal? In practice it is probably something in between. Why? Because empirical procedures enforce a *one-gap* representation inside a general state-space system. Output is the sum of a cyclical gap c and a potential level τ . This design compresses several theoretically distinct gaps and several counterfactual notions of “potential” into (c_t, τ_t) . The interpretation of the one reported gap and counterfactual become muddled. The gaps capture deviations from frictionless benchmarks but multiple frictions exist; nominal rigidities and real rigidities such as search and matching frictions, market power, imperfect information among many. All of these distinct gaps can represent the deviation from a counterfactual economy with some subset of frictions turned of and all these gaps can have distinct implications. Interpreting the one estimated gap is impossible unless the model specification is such that exactly one specific gap is extracted. Only then is identification and interpretation of the gap possible.

Setup: state-space with a single gap. Conventional methods for empirical output-gap estimation can be written in the state-space form

$$s_t = \Phi s_{t-1} + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \Sigma_\xi), \quad (1)$$

$$x_t = H s_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon), \quad (2)$$

with $s_t \in \mathbb{R}^{n_s}$ unobserved states and $x_t \in \mathbb{R}^{n_x}$ observables (e.g., $x_t = (y_t, u_t, \pi_t, E_t \pi_{t+1}, \dots)$). Conventional implementations embed the identity

$$y_t = \tau_t + c_t + \varepsilon_t^y,$$

where τ_t (trend/potential) and c_t (cycle/gap) are components of s_t , and the output row of H decomposes as $y_t = h'_{y,\tau} s_t + h'_{y,c} s_t + \varepsilon_{y,t}$ with $h'_{y,\tau} = h'_{y,c} = 1$. This state space formulation includes, among many popular choices for output gap measurement, the HP-filter, band-pass filters, Beveridge-Nelson decompositions, Blanchard-Quah decompositions, structural time series and unobserved components models (Harvey, 1985, Clark, 1987, Kuttner, 1994, Jarociński and Lenza, 2018, Hasenzagl *et al.*, 2022), production function approaches and structural VARs. **(CHECK ALL THIS)**

Result: the gap is a mixture of structural gaps. Let $\{\tau_t^k\}_{k=1}^K$ denote economically meaningful counterfactual output concepts (flex-price/natural, efficient, financial-frictionless, trend/steady state, *etc.*). Define the corresponding structural gaps

$$c_t^k \equiv y_t - \tau_t^k, \quad c_t \equiv (c_t^1, \dots, c_t^K)'$$

Assume observables are generated by these gaps:

$$x_t = \Lambda c_t + m_t, \quad \mathbb{E}[c_t m_t'] = 0, \quad (3)$$

where m_t collects dynamics unrelated to the gaps.

A conventional *one-gap* specification instead *misspecifies* the observation system as

$$x_t = \lambda g_t^* + \tilde{m}_t, \quad \lambda_y = 1, \quad y_t = \tau_t^* + g_t^*,$$

and estimates (1)–(2) with a single scalar gap g_t^* and a single potential τ_t^* (plus measurement blocks). When $K > 1$ in the true system (3), the estimated gap g_t^* is not any one structural gap but a *fixed linear combination* of them:

$$g_t^* = w' c_t = \sum_{k=1}^K w_k c_t^k,$$

with some weights $w \in \mathbb{R}^K$.

Implication. Only in the case $w_j \approx 1$ and $w_{-j} \approx 0$ (weight one on a single structural gap c_t^j , zero on the rest) does g_t^* coincide with an interpretable structural gap. Otherwise, the reported “gap” and “potential” are *mixtures* of a range of relevant gaps and identification and interpretation is impossible.

Example: One-gap ambiguity in two nearly identical systems. In a reduced-form observation system where a single common gap g_t drives both real activity and inflationary pressure, the observables can be written as

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ E_t \pi_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ \delta_u & 0 \\ \delta_\pi & 1 \\ \delta_{E\pi} & 1 \end{pmatrix}}_{\text{common states loadings}} \begin{pmatrix} g_t \\ \mu_t^\pi \end{pmatrix} + \underbrace{\begin{pmatrix} \mu_t^y \\ \mu_t^u \\ 0 \\ \mu_t^{E\pi} \end{pmatrix}}_{\text{idiosyncratic persistencies}} + \underbrace{\begin{pmatrix} \Psi_t^y \\ \Psi_t^u \\ \Psi_t^\pi \\ \Psi_t^{E\pi} \end{pmatrix}}_{\text{idiosyncratic fluctuations}}, \quad (4)$$

where μ and Ψ absorb permanent and transitory idiosyncratic movements, respectively. This specification—call it *model 1*—identifies g_t from the comovement of real activity and nominal variables. This formulation is identical to the simplest model considered in Hasenzagl *et al.* (2022), but excluding oil prices. They argue that this model broadly captures the common view of the economy held by policy makers with g_t capturing the business cycle, but in a gap notion it is unclear which *structural* gap g_t represents. We can consider a tiny extension—*model 2*—adds a common transitory nominal cycle $\Psi_t^{\pi, E\pi}$ that loads on π_t and $E_t \pi_{t+1}$ but is not linked to the real side of the economy:

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ E_t \pi_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ \delta_u & 0 & 0 \\ \delta_\pi & 1 & 1 \\ \delta_{E\pi} & 1 & \gamma_{E\pi} \end{pmatrix}}_{\text{common states loadings}} \begin{pmatrix} g_t \\ \mu_t^\pi \\ \Psi_t^{\pi, E\pi} \end{pmatrix} + \underbrace{\begin{pmatrix} \mu_t^y \\ \mu_t^u \\ 0 \\ \mu_t^{E\pi} \end{pmatrix}}_{\text{idiosyncratic persistencies}} + \underbrace{\begin{pmatrix} \Psi_t^y \\ \Psi_t^u \\ \Psi_t^\pi \\ \Psi_t^{E\pi} \end{pmatrix}}_{\text{idiosyncratic fluctuations}}, \quad (5)$$

This minor change can flip the interpretation of g_t : in (4) the gap may predominantly explain nominal fluctuations, while in (5) nominal comovement can be absorbed by the added nominal factor so the estimated g_t shifts to explain real-side fluctuations. The same “one-gap” label thus supports two different narratives because g_t is a mixture of relevant gaps with weights adjusting to fit the observation system. The estimated cyclical components illustrate this point below.

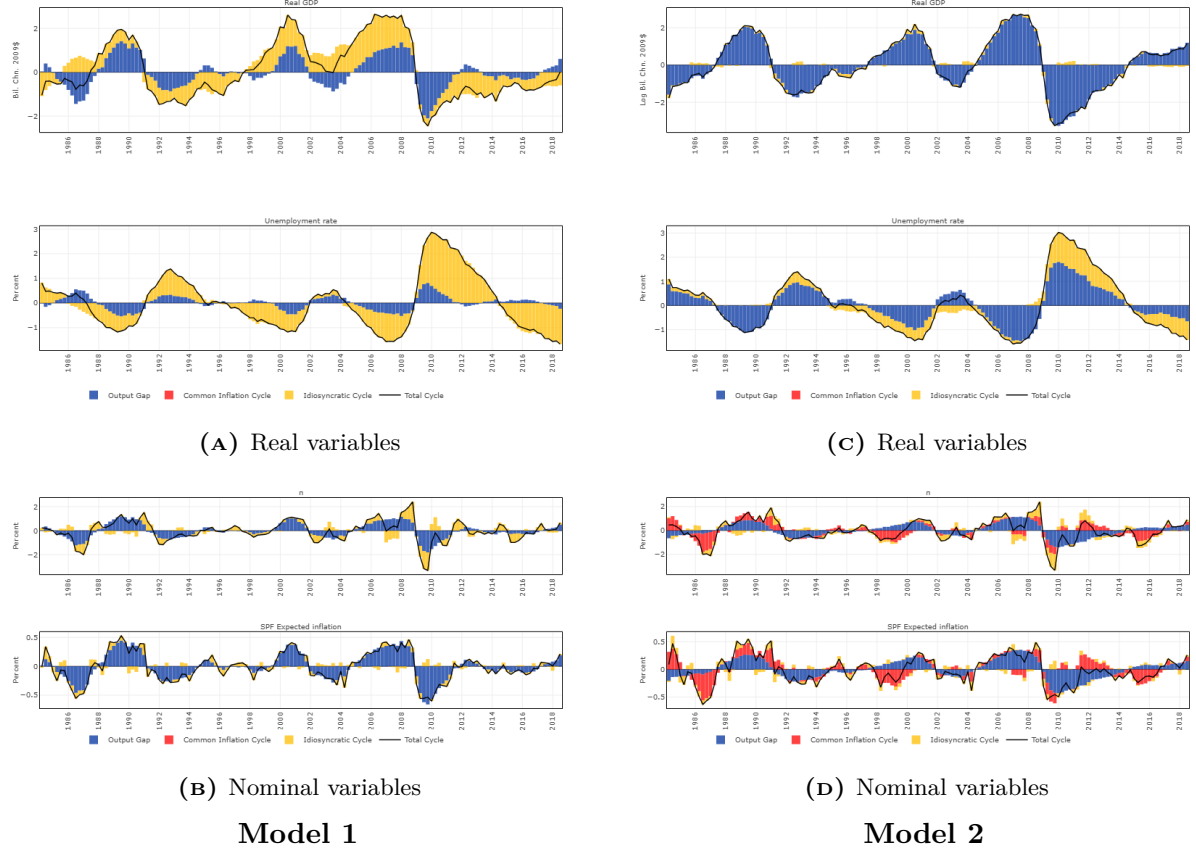


FIGURE 1. Cycle decompositions

The Textbook NK model as a framework. We use the textbook New Keynesian (NK) model (e.g. Galí, 2015) as a minimal structure that clarifies the ambiguity above. The model features two core frictions: (i) *nominal rigidity* (sticky prices) and (ii) *real rigidity* (market power/monopolistic distortion). These two frictions generate two economically meaningful counterfactual output levels and, hence, two relevant policy gaps: the *flex-price (natural)* allocation y_t^{flex} obtained by removing nominal rigidity, and the *efficient* allocation y_t^{eff} obtained by removing market power. The gaps are:

$$c_t^{flex} \equiv y_t - y_t^{flex}, \quad c_t^{eff} \equiv y_t - y_t^{eff}.$$

In the model the first gap, c_t^{flex} , is the inflation-relevant gap (price stability), while the second, c_t^{eff} , is the resource-utilization/welfare-relevant gap (efficient use of labor).

This mapping slots directly into the reduced-form systems in (4)–(5). When nominal comovement *must* load on a single gap (model (4)), the estimated one-gap g_t is pushed toward the inflation-relevant direction and behaves like c_t^{flex} . When a separate common nominal factor $\Psi_t^{\pi, E\pi}$ is admitted (model (5)), nominal comovement can be absorbed outside the gap, so the estimated g_t reorients toward real-side variation and aligns more closely with c_t^{eff} . In general, the single statistical gap is a mixture

$$g_t = w_{flex} c_t^{flex} + w_{eff} c_t^{eff},$$

with dynamic weights (w_{flex}, w_{eff}) determined by the empirical model formulation and signals from observations. The apparent “flip” in what g_t explains across (4) and (5) is therefore not a paradox but the predictable re-weighting of a one-dimensional estimate when the system allows nominal movements to be captured apart from the gap. This shows how a one-gap system is informative only when the specification *forces* the weights to pick out the intended structural gap. Theoretical structure is needed to convincingly argue that an actual true and relevant gap is extracted. A stationary unobserved cycle loading with unity on output is not an output gap unless theoretical structure is imposed to force it as such.

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